

ABRAR SHAIK

Python Developer

San Francisco, CA Open for relocation to Seattle, WA | Plano, TX | Chicago, IL | New York, NY
630-489-8943 | abrar.sh122144@gmail.com | <https://www.linkedin.com/in/abrar-s-354b2836a/>

SUMMARY

Python Developer with 5+ years of experience delivering scalable APIs, LLM inference systems, and cloud-native architectures across Perplexity AI, Meta, and Accenture. Proven expertise in Python, FastAPI, Flask, gRPC, MLOps, and multi-cloud (AWS/GCP/Azure), with hands-on work in LLM integration, observability, and automation. Adept at cross-functional collaboration, mentoring, and driving high-performance, fault-tolerant systems that improve user engagement, reliability, and business impact.

PROFESSIONAL EXPERIENCE

Perplexity AI – pplx-API & Comet / Model Integration Project

January 2024 – Present

Backend Software Engineer (Python)

San Francisco, CA

- Developed pplx-API backend powering low-latency LLM inference, reducing response times 70% while seamlessly scaling to millions of Pro/Max production queries.
- Integrated LLMs including GPT-4.1, Claude, LLaMA, and Mistral supporting JSON/Regex structured outputs, doubling developer adoption and delivering enterprise-grade reliability across diverse environments.
- Contributed to launch of Comet AI browser functionalities and Perplexity Max platform features, driving improved engagement, stronger user retention, and measurable customer growth.
- Built scalable inference-ready Python APIs leveraging FastAPI, Flask, and gRPC while optimizing model performance using TensorRT-LLM across AWS GPU-based clusters.
- Designed, containerized, and deployed highly-available microservice ecosystems using Docker, Kubernetes, and auto-scaling strategies ensuring reliable fault-tolerant cloud-native backend infrastructures.
- Implemented automated MLOps pipelines using MLflow, Weights & Biases, and proprietary model stores, streamlining rollout/rollback processes and monitoring critical LLM model deployments.
- Developed structured response generation features powering RAG, summarization, and automation workflows, enabling advanced client-side application integrations at enterprise production scale.
- Enhanced observability stack by integrating Prometheus, Grafana, and ELK/Loki, delivering advanced diagnostics, system monitoring, and SLA-compliant real-time metrics visibility.
- Automated deployment lifecycles with CI/CD GitHub Actions and Jenkins, reinforcing quality with PyTest-driven tests and optimized throughput via caching and dynamic batching.
- Collaborated within Agile/Scrum cross-functional teams, mentoring junior developers, ensuring continuous code quality improvements, and demonstrating strong technical communication and problem-solving capabilities.

Meta

February 2023 – December 2023

Software Engineer

San Francisco, CA

- Developed and fine-tuned LLaMA 2/3 large language models using PyTorch for generative ad copy, improving Meta Ads click-through rates by 11% globally.
- Built and deployed vision diffusion models including Emu for image background generation and video animation, increasing ad engagement 13% while accelerating creative delivery processes.
- Implemented automated model retraining workflows seamlessly integrated into CI/CD pipelines via GitHub Actions and Jenkins, ensuring rapid iteration, reliability, and reproducibility in deployment cycles.
- Engineered real-time ad recommendation models (GEM) leveraging deep neural networks and reinforcement learning, generating 7.6% measurable lift in conversions across 4M+ advertiser base.
- Designed scalable inference services with Docker, Kubernetes, and Meta MTIA accelerators, reducing latency by 40% while sustaining over 10M+ daily requests.
- Optimized multi-modal AI pipelines for Reels and Feed personalization using transformers across text-image models, boosting user engagement and time spent up to 8%.
- Deployed optimized generative models with ONNX, TorchScript, and quantization, maintaining sub-100ms inference latency consistently across globally distributed Meta data centers.
- Built scalable database integration pipelines with Spark SQL and Presto, enabling fast, distributed ad platform data retrieval for ML-powered personalization services.

- Orchestrated A/B testing and canary rollout of generative models via FBLeaer Flow ML Ops, verifying over 5% incremental production ad performance lift.
- Partnered with Product, Data Science, and Responsible AI teams to embed invisible watermarks, safety filters, and brand compliance across generative AI-driven creatives.
- Designed Spark, Presto, and Scuba pipelines processing petabyte-scale ad interaction logs, enabling automated retraining cycles and actionable business intelligence for marketing optimization.
- Integrated text-to-image and text-to-video generation into Ads Manager platform, accelerating creative workflows by 60% and scaling AI augmentation access to 1M+ advertisers.

Accenture

May 2019 – June 2022

Python Application Developer

India

- Developed and deployed Python-based automation workflows that streamlined enterprise reporting and analytics, reducing manual processing time by 50% and improving client decision-making speed.
- Built scalable ETL pipelines to integrate multi-source business data (ERP, CRM, finance systems), enhancing data accuracy and supporting real-time dashboards for global clients.
- Designed and maintained backend services in Python (Flask, FastAPI) with strong integration to SQL/NoSQL databases, implementing data validation, monitoring, and fault tolerance.
- Leveraged AWS (EC2, S3, Lambda, RDS) and Azure services to deliver cloud-native solutions, while implementing CI/CD (Jenkins, GitHub Actions) and test automation (PyTest) for reliability.
- Collaborated in Agile/Scrum teams, applying problem-solving, communication, and stakeholder management skills, while mentoring junior developers and adopting best practices in MLOps, observability (Prometheus, Grafana), and containerization (Docker, Kubernetes).

TECHNICAL SKILLS

Programming & Scripting: Python (FastAPI, Flask, Django, gRPC), JavaScript/TypeScript (basic integration), Bash, SQL, NoSQL

AI/ML & Data Engineering: Large Language Models (LLMs – GPT-4.1, Claude, LLaMA, Mistral), Hugging Face, PyTorch, TensorRT-LLM, Transformers, RAG (Retrieval-Augmented Generation), Semantic Search, Vector Databases (FAISS, Pinecone, Weaviate, Milvus), Recommendation Systems, NLP (Sentiment Analysis, Text Classification), Data Ingestion & ETL Pipelines, Real-time Stream Processing (Kafka, Pub/Sub)

MLOps & Model Lifecycle: MLflow, Weights & Biases (W&B), Airflow, Model Stores, Automated Rollout & Rollback, Model Monitoring, SageMaker, KubeFlow (trending 2025), Feature Stores

Cloud Platforms & Services: AWS (EC2, S3, Lambda, SageMaker, DynamoDB, RDS, GPU Clusters), GCP (BigQuery, Pub/Sub), Azure (App Services, Functions, Storage), Cloud-Native Architectures, Multi-Cloud Deployments

DevOps, Infrastructure & Observability: Docker, Kubernetes (EKS, GKE, AKS), Terraform (IaC – trending 2025), Helm, Auto-scaling Systems, CI/CD Pipelines (GitHub Actions, Jenkins), Monitoring & Logging (Prometheus, Grafana, ELK, Loki), OpenTelemetry (trending 2025 for observability), Fault-tolerant High Availability Systems

Databases & Storage: PostgreSQL, MySQL, DynamoDB, Redis (caching & performance optimization), Elasticsearch, Time-series Databases (InfluxDB, Prometheus TSDB)

Testing & Performance Optimization: PyTest, Unit & Integration Testing, Load/Stress Testing, Performance Optimization (Batching, Caching, Latency Reduction), A/B Testing

Soft Skills & Leadership: Agile/Scrum Methodologies, Cross-functional Collaboration, Mentoring & Coaching, Communication & Stakeholder Management, Problem Solving, Code Review & Quality Improvement, Project Delivery & Ownership

EDUCATION

Master of Science in Information System

Trine University, USA